VHUTHELU RESOURCES

Secure Hybrid Network Design & Simulation

Task 1 – Implementation & Technical Documentation

Item	Details
Organisation	Vhuthelu Resources
Project	Secure Hybrid Office and Remote Network
Simulation Tool	Cisco Packet Tracer
Primary Devices	Cisco ASA 5505, Cisco 2911, Cisco 2960
Network Model	Hybrid office + remote/home connectivity
Purpose	WIL / Industry Project – Task 1

1. Project Overview

The objective of this task was to design and simulate a secure hybrid network for Vhuthelu Resources that supports office-based and remote/home users. The design separates employees, servers, guests and management devices into different VLANs, provides inter-VLAN routing, applies access-control rules, provides controlled WAN connectivity, and keeps remote users separated from the protected corporate LAN.

2. User and Network Requirements

- 1 Manager.
- 1 IT Team Lead and 6 Developers.
- A company server containing important organisational information.
- Guest devices requiring Internet access without internal access.
- Two remote/home users represented by two laptops.
- VLANs for Employees, Servers, Guests and Management.
- Routing and access-control rules.
- Security and connectivity testing and documentation.

3. Network Architecture

Office path: Simulated Internet/ISP → ASA firewall → R1 router → SW1 core/access switch → VLAN endpoints.
Remote path: Remote laptops → SW-HOME → HOME-ROUTER → simulated ISP/WAN.

The ASA 5505 is the perimeter firewall, R1 performs router-on-a-stick inter-VLAN routing, and SW1 provides VLAN segmentation. The remote network is separate from the office VLANs.

4. Devices and Roles

Device	Role	Connection
FW1-VHUTHELU (ASA 5505)	Perimeter firewall, NAT, WAN boundary	Outside to ISP; inside to R1
R1-VHUTHELU (2911)	Inter-VLAN routing/default gateways	G0/0 trunk to SW1; G0/1 to ASA
SW1-CORE (2960)	VLAN/access switching	Gi0/1 trunk to R1
SRV1-VHUTHELU	Company server	VLAN 20
PC-MANAGER	Manager workstation	VLAN 40
PC-IT-LEAD	IT Team Lead	VLAN 10
PC-DEV1–DEV6	Developer workstations	VLAN 10
PC-GUEST1–GUEST2	Guest devices	VLAN 30
SW-HOME (2960)	Remote/home switch	Laptops to HOME-ROUTER
HOME-ROUTER (2911)	Remote/home gateway	Home LAN to WAN
LAPTOP-REMOTE1/2	Remote users	Home LAN
ISP-ROUTER (2911)	Simulated ISP/WAN	Office and remote WAN paths

5. VLAN and IP Addressing Plan

Network	Purpose	Subnet	Gateway	Addressing
VLAN 10 – EMPLOYEES	IT Lead + Developers	192.168.10.0/24	192.168.10.1	DHCP .100–.200
VLAN 20 – SERVERS	Company server	192.168.20.0/24	192.168.20.1	Static
VLAN 30 – GUESTS	Guest devices	192.168.30.0/24	192.168.30.1	DHCP .100–.200
VLAN 40 – MANAGEMENT	Manager	192.168.40.0/24	192.168.40.1	Static
Home LAN	Remote users	192.168.50.0/24	192.168.50.1	DHCP
ASA ↔ R1	Firewall transit	10.0.0.0/30	ASA .1 / R1 .2	Static
ASA ↔ ISP	Office WAN	192.168.100.0/24	ASA .1 / ISP .254	Static
ISP ↔ HOME-ROUTER	Remote WAN	203.0.113.0/30	ISP .1 / Home .2	Static

6. Endpoint Addressing

Endpoint	Address	Gateway
PC-MANAGER	192.168.40.10/24	192.168.40.1
SRV1-VHUTHELU	192.168.20.10/24	192.168.20.1
PC-IT-LEAD	DHCP	192.168.10.1
PC-DEV1–DEV6	DHCP	192.168.10.1
PC-GUEST1–GUEST2	DHCP	192.168.30.1
LAPTOP-REMOTE1/2	DHCP	192.168.50.1

7. How Addressing Was Designed

Each functional VLAN uses a /24 subnet for simplicity and manageability. The third octet identifies the function: 10 Employees, 20 Servers, 30 Guests and 40 Management. The remote/home network uses 192.168.50.0/24. Point-to-point transit links use /30 networks because only two routed interfaces are required.

Static addresses were used for important infrastructure and endpoints such as the server and Manager PC. DHCP was used for employee, guest and home-user endpoints.

8. Switching and Routing

SW1-CORE was configured with VLANs 10, 20, 30 and 40. Employee ports were assigned to VLAN 10, the server to VLAN 20, guest ports to VLAN 30 and the Manager to VLAN 40. SW1 Gi0/1 to R1 G0/0 was configured as an 802.1Q trunk.

R1 uses G0/0.10, G0/0.20, G0/0.30 and G0/0.40 as router-on-a-stick subinterfaces. Each provides the default gateway for its VLAN. R1's default route points to the ASA at 10.0.0.1.

9. Firewall, NAT and WAN

The ASA 5505 was positioned at the perimeter. Inside is 10.0.0.1/30 and outside is 192.168.100.1/24. Static routes were added to the four internal VLANs through R1 at 10.0.0.2. Dynamic PAT/NAT was configured for internal VLANs to provide simulated WAN access.

The ISP router simulates the external network. The remote WAN uses 203.0.113.0/30 and the remote LAN uses 192.168.50.0/24.

10. Security Controls

Control	Implementation	Expected Result
Guest isolation	GUEST-ISOLATION ACL on VLAN 30	Guests cannot reach Employees, Servers or Management
Employee restriction	EMPLOYEE-RESTRICTION ACL on VLAN 10	Employees cannot reach Management
Server protection	Dedicated VLAN 20; guest access blocked; no direct Internet exposure	Reduced unauthorised access
Perimeter security	ASA firewall + NAT + routing controls	Internal addressing protected at WAN boundary
Remote separation	Separate 192.168.50.0/24 network	Remote users are not office VLAN members
ICMP inspection	ASA global policy	Required return ICMP traffic works through firewall

11. Key ACL Rules

- Guest → Employees: DENY
- Guest → Servers: DENY
- Guest → Management: DENY
- Guest → other permitted destinations: ALLOW as required for Internet access
- Employees → Management: DENY
- Employees → other required destinations: ALLOW

12. Testing and Results

Test	Expected / Observed Result	Purpose
Employee → Server	Allowed	Required server access
Manager → Server	Allowed	Management server access
Guest → Server	Blocked	Guest isolation
Guest → Employee	Blocked	Guest isolation

Guest → Management	Blocked	Guest isolation
Employee → Management	Blocked	Management protection
Employee → ISP/WAN	Allowed	NAT/WAN validation
Guest → ISP/WAN	Allowed	Guest Internet access
Remote → Home Gateway	Allowed	Remote LAN validation
Remote → Corporate Network	Blocked	Remote network not automatically trusted
ASA ↔ R1	Allowed	Firewall/router transit
ASA ↔ ISP	Allowed	Perimeter WAN connectivity

13. VPN / Remote Access Limitation

An IPsec/IKEv1 remote-access design was investigated. The ASA 5505 accepted the IKEv1 policy, IPsec transform-set, crypto-map and tunnel-group configuration. However, the selected HOME-ROUTER in Packet Tracer did not expose the required IKE/IPsec commands; its crypto options only exposed long-term key operations. Therefore a complete working ASA-to-home IPsec tunnel could not be demonstrated with the selected simulated devices.

The VPN configuration was disabled/removed from active operation rather than presenting an incomplete tunnel as working. The final simulation demonstrates the security boundary: remote users are separated from the corporate LAN and are not automatically trusted. In production, authorised remote users should use an approved authenticated and encrypted VPN solution.

Note: a small amount of unused crypto configuration may remain in the ASA running configuration because Packet Tracer did not allow every crypto-map subcommand to be removed cleanly. IKEv1 was disabled and the crypto map was not applied to the outside interface, so the VPN is not being presented as active.

14. Other Limitations

- Packet Tracer does not implement every feature/behaviour of production Cisco platforms.
- The Internet/WAN is simulated with an ISP router.
- Detailed service-by-service server ACLs were kept outside the one-week task scope.
- The IT Team Lead remains in VLAN 10; VLAN 40 is reserved for the Manager and restricted management traffic.

15. Key Notes for Presentation / Demonstration

Why VLANs? To separate user groups and reduce lateral movement.

Why VLAN 20? The server contains important information, so it has a dedicated protected segment.

Why VLAN 40? Management traffic is separated from normal employee traffic.

Why isolate guests? Guests need Internet access but should not access company systems.

Why R1? It provides inter-VLAN routing and default gateways.

Why ASA? It is the perimeter security boundary and performs NAT.

Why DHCP? It simplifies endpoint deployment.

Why static IPs? Important infrastructure has predictable addresses.

Why two remote laptops? They represent separate remote users/home locations.

Why no working VPN? The selected Packet Tracer home-router image did not support the required IKE/IPsec commands; this was documented as a simulator limitation.

16. Evidence / Screenshots to Include

- Final Packet Tracer topology.
- SW1 VLAN and trunk evidence.
- R1 routing and interface evidence.
- ASA interface, route and NAT evidence.
- Employee → Server success.
- Manager → Server success.
- Guest → Server blocked.
- Guest → Employee blocked.
- Employee → Management blocked.
- Remote laptop IP and home-gateway success.
- Remote → corporate network blocked.
- VPN command limitation evidence, if required.

17. Conclusion

The simulated network meets the main Task 1 objectives by providing a segmented hybrid network with controlled routing, perimeter security, guest isolation, management protection, protected server placement and separated remote connectivity. The design provides a practical foundation for a production hybrid network while clearly identifying Packet Tracer limitations.

End of Task 1 Technical Documentation